SQL DB

what is DataBase –

It is a collection of Data in a format that can be easily Accessed.

SQL V/S NoSQL –

SQL –

Relational DB (Data stored in tables)

eg- MySQL, Oracle, PostgreSQL etc

NoSQL –

- Non Relational Database (Data stored in Document / Key - Val / graphs etc)

eg- MongoDB, Cassandra, Neo4j etc.

SQL (Structured Query language)

SQL is a programming lang. used to interact with Relational DB.

our 1st DB

CREATE DATABASE College;

DROP DATABASE College; — For Deleting DB.

USE college; — For creating Tables / working on DB.

Syntax -

```
CREATE TABLE table_name (
  column-name 1 datatype constraint,
  column-name 2 datatype constraint,
  column-name 3 datatype constraint,
);
```

Eg-

```
CREATE DATABASE College ;
USE college;
CREATE TABLE student (
  rollno INT,
  name VARCHAR (30),
  age INT
);

INSERT INTO Student
VALUES
(101, "adam", 12),
(102, "bob", 14);

SELECT * FROM student;
```

Database Queries.

CREATE DATABASE db_name;
CREATE DATABASE IF NOT EXISTS db_name;

DROP DATABASE db_name;
DROP DATABASE IF EXISTS db_name;

SHOW DATABASE;
SHOW TABLES;

Table Queries.

Create - For creating Table
insert - For inserting Data into Table
update - For updating Data
Alter - For Altering schema, changing Columns.
Truncate - Delete all Data & keep empty
Delete - Deleting Table.

Constraints - Rules For Data in the Table

NOT NULL - Columns cannot Have a null Value
UNIQUE - all Values in column are Different
DEFAULT - Sets the Default value of a Column
CHECK - it can limit the values allowed in Column

PRIMARY KEY-
makes a column unique & not null
but used only for one.

```
CREATE TABLE temp (
  id int not null,
     PRIMARY KEY (id)
);
```

FOREIGN KEY-
Prevent Actions that would destroy
links between tables

```
CREATE TABLE temp (
   cust_id int,
   FOREIGN KEY (cust_id) references customer (id)
);
```

What are keys?
keys are special coloums in the Table

Primary key-
It is a coloumn (or set of coloumns) in a Table that uniquely identifies each row (a unique id)
There is only 1 PK & it should be NOT NULL.

Foreign key-
A foregin key is a coloumn (or set of coloums) in a Table that refers to the primary key

in another Table FKs can Have duplicate & Null values.
there can be multiple FKs.

Table Queries
insert Into Table

Syntax-

```
INSERT INTO table_name
(colname 1, colname 2);
VALUES
(col1_v1, col2_v1),
(col1_v2), (col2_v2);
```

Eg-

```
INSERT INTO user
(id, age, name, email, followers, following)
VALUES
(1, 14, "adam", "adm@yahoo.in", 123, 145),
(2, 15, "bob", "bob@bob-in", 157, 200);
```

Select Command
select & show Data from the DB

Syntax

```
SELECT col1, col2 FROM table_name;
```

Syntax (to show all)

SELECT * FROM table-name;

where clause
To define conditions

SELECT col1, col2 FROM table_name
WHERE conditions;

Eg -
SELECT name, followers
FROM user
WHERE followers >= 200;

Another operators -

Arithmetic operators : +, -, *, /, %
Comparison operators : =, !=, >, <, >=, <=
logical operators : AND, OR, NOT, IN, BETWEEN
Bitwise operators : &, |

Frequently used operators.

AND (to check for both conditions to be true)
OR (to check for one of the conditions to be true)
BETWEEN (selects for a given Range)
IN (matches any value in the list)
NOT (to negate the given condition)

Order by Clause -
To Sort in Ascending (ASC) or descending order (DESC)

SELECT Col1, Col2 FROM table_name
ORDERBY Col-name (s) ASC;

Limit Clause -
Sets an upper limit on number of (tuples) rows to be Returned.

SELECT Col1, Col2 FROM table-name
LIMIT number -

Aggregate Functions -
Aggregate Functions Perform a Calculation on a set of values, and return a single value.

- COUNT()
  MAX()
  MIN()
  SUM()
  AVG()

Eg -

SELECT max(marks)
FROM Student;

Group by Clause -

Groups rows that Have the Same values into summary rows.
It Collects Data From multiple records & groups the Result by one or more Column.

```
SELECT col1, col2
FROM table_name
GROUP BY col_name(s);
```

* Generally we use group by with some Aggregation function.

Having Clause -

Similar To where i.e. applies Some conditions on rows But it is used when we want to apply any condition after grouping.

```
SELECT col1, col2
FROM table_name
GROUP BY col_name(s)
HAVING condition;
```

- where is For the table, HAVING is For a group
- Grouping is necessary For HAVING.

Genral order -

```
SELECT Coloumn (s)
FROM table-name
WHERE Condition
GROUP BY Coloumn(s)
HAVING Condition
ORDER BY Col (s) ASC;
```

Table Queries

UPDATE (To update Existing Row)

```
UPDATE table-name
SET Col1 = val1, Col2 = val2
WHERE Condition;
```

DELETE (To Delete existing row)

```
DELETE FROM table-name
WHERE Condition;
```

Ex -

```
DELETE FROM user
WHERE age = 14;
```

ALTER (To change the Schema)

Add Coloumn
ALTER TABLE table_name
ADD COLOUMN column_name datatype constraint;

DROP Coloumn
ALTER TABLE table_name
DROP COLUMN coloumn_name;

RENAME TABLE
ALTER TABLE table_name
RENAME TO new_table_name;

TRUNCATE (to Delete table's Data)

TRUNCATE TABLE table_name;